

Mebrahtu F. Weldeghebriel

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RESEARCH INTERESTS

Low-temperature and isotope geochemistry · paleoclimatology · ancient seawater chemistry · long-term carbon cycle · evaporites, brines, carbonates and dolostones · critical mineral resources in sedimentary basins · ice-core geochemistry · development of in situ analytical methods.

ACADEMIC POSITIONS

Gordon and Betty Moore Postdoctoral Research Fellow, Princeton University	Sep 2026 – Present
NSF Postdoctoral Research Fellow, Princeton University	Sep 2024 – Sep 2026
Harry H. Hess Postdoctoral Research Fellow, Princeton University	Sep 2022 – Sep 2024
Graduate Research and Teaching Assistant, Binghamton University	Aug 2016 – Jul 2022
Lecturer and Teaching Assistant, Eritrea Institute of Technology	Sep 2011 – Jun 2015

EDUCATION

Binghamton University, State University of New York, Binghamton, NY Aug 2016 – Jul 2022

PhD, Geological Sciences

Dissertation: *Secular variability in major, minor, and trace element composition of Phanerozoic and Neoproterozoic seawater: evidence from fluid inclusions in marine halite*. Advisor: Dr. Tim Lowenstein, Distinguished Professor.

Eritrea Institute of Technology (EIT), Asmara, Eritrea Sep 2006 – Jul 2011

BSc, Geology

Dissertation: *Sedimentology and lithostratigraphic characteristics of the evaporite succession in the Colluli area, Danakil Depression, East African Rift, Eritrea*. Advisor: Dr. Nageshwar Dubey, Professor.

HONORS, FELLOWSHIPS, AND AWARDS

- NSF Office of Polar Programs Postdoctoral Research Fellowship ([OPP 2420291](#)), \$359,487 (2024–2026), Principal Investigator.
- Gordon and Betty Moore Foundation Postdoctoral Fellowship, \$60,000 (2026–2027).
- Harry H. Hess Postdoctoral Research Fellowship, Princeton University, \$147,838 (2022–2024).
- Graduate research grants: Geological Society of America MGPV Division, \$2,500 ([2019](#)); Society of Economic Geologists Foundation, \$4,700 ([2018](#)); American Association of Petroleum Geologists Foundation, \$3,000 ([2018](#)).
- Graduate Student Excellence Award in Research, Binghamton University (2021).
- Graduate Student Research Award for excellence in research proposal, GSA MGPV Division (2019).
- Research Assistantship Scholarship Award, Binghamton University (2021–2022).
- Teaching Assistantship Scholarship Award, Binghamton University (2016–2021).
- Provost Doctoral Summer Fellowship, Binghamton University (2016–2020).
- Türkiye Scholarship, Dokuz Eylül University, Turkey (2015).
- Outstanding Student and Gold Medal Winner, College of Science, Eritrea Institute of Technology (2011).

PUBLICATIONS

Weldeghebriel, M. F., Lowenstein, T. K., 2023. Seafloor hydrothermal systems control seawater [Li⁺]: evidence from fluid inclusions. *Science Advances* 9, eadfl605. <https://doi.org/10.1126/sciadv.adfl605>

Weldeghebriel, M. F., Lowenstein, T. K., Xia, Z., Li, W., 2023. Plate tectonic control of strontium concentration in Phanerozoic and Neoproterozoic seawater: evidence from fluid inclusions in marine halite. *Geochimica et Cosmochimica Acta* 346, 165–179. <https://doi.org/10.1016/j.gca.2023.02.009>

Weldeghebriel, M. F., Lowenstein, T. K., García-Veigas, J., Cendón, D. I., 2022. [Ca²⁺] and [SO₄²⁻] in Phanerozoic and terminal Proterozoic seawater from fluid inclusions in halite: the significance of Ca-SO₄ crossover points. *Earth and Planetary Science Letters* 594, 117712. <https://doi.org/10.1016/j.epsl.2022.117712>

- Weldeghebriel, M. F.**, Lowenstein, T. K., García-Veigas, J., Collins, D., Sendula, E., Bodnar, R. J., Graney, J. R., Cendón, D. I., Lensky, N. G., Mor, Z., Sirota, I., 2020. Combined LA-ICP-MS and cryo-SEM-EDS: an improved technique for quantitative analysis of major, minor, and trace elements in fluid inclusions in halite. *Chemical Geology* 551, 119762. <https://doi.org/10.1016/j.chemgeo.2020.119762>
- Yang, Y., Liu, Y., Pogge von Strandmann, P. A., Jin, Z., Galy, A., Ye, C., **Weldeghebriel, M. F.**, Yan, Z., He, J., Gou, L. F., Deng, L., 2026. Decoupling of Neogene seawater lithium isotopes from uplift-driven weathering. *Nature Communications* 17, 6524. <https://doi.org/10.1038/s41467-026-71407-x>
- Xia, Z., Li, S., Hu, Z., Bialik, O., Chen, T., **Weldeghebriel, M. F.**, ... Li, W., 2024. The evolution of Earth's surficial Mg cycle over the past 2 billion years. *Science Advances* 10, eadj5474. <https://doi.org/10.1126/sciadv.adj5474>
- Li, Y., Wang, Y. Y., Murphy, J. G., **Weldeghebriel, M. F.**, Xiao, Y., 2024. Evaluating evaporites as potential archives of lake and seawater lithium isotopic ratio: an experimental study using various natural saline fluids. *Chemical Geology* 122326. <https://doi.org/10.1016/j.chemgeo.2024.122326>
- Lowenstein, T. K., **Weldeghebriel, M. F.**, Sirota, I., Eyal, H., Mor, M., Lensky, N. L., 2021. Criteria for the recognition of clastic halite: the modern Dead Sea shoreline. *Sedimentology* 68, 2253–2269. <https://doi.org/10.1111/sed.12907>
- Ajrrough, S., Boutarouine, H., Lowenstein, T. K., **Weldeghebriel, M. F.**, Xia, Z., El Arabi, E. H., 2024. Evaporites and red beds of a syn-rift Atlantic series: evidence of continental depositional environment (Berrechid sub-basin, Morocco). *Journal of Sedimentary Research* 94, 750–767. <https://doi.org/10.2110/jsr.2023.025>
- Sendula, E., Gill, B. C., Rimstidt, J. D., Lowenstein, T. K., **Weldeghebriel, M. F.**, García-Veigas, J., Bodnar, R. J., 2022. Redox conditions in Late Permian seawater based on trace metal ratios in fluid inclusions in halite from the Polish Zechstein Basin. *Chemical Geology* 596, 120794. <https://doi.org/10.1016/j.chemgeo.2022.120794>

MANUSCRIPTS IN REVIEW AND IN PREPARATION

- Weldeghebriel, M. F.**, et al. A 550-million-year record of the concentration and isotopic composition of lithium in seawater. *Science Advances*, in review.
- Guillerm, E., et al. (incl. **Weldeghebriel, M. F.**) High-fidelity reconstructions of element concentrations in halite fluid inclusions with Brillouin–LA-ICP-MS. *Chemical Geology*, in review.
- Weldeghebriel, M. F.**, et al. Secular variation in seawater lithium and the origin of lithium-rich basinal brines. In preparation.
- Weldeghebriel, M. F.**, et al. Conservative behavior of Li during extreme evaporation of seawater — an experimental study. In preparation.

SELECTED CONFERENCE ABSTRACTS [#]Denotes invited. ^{*}Denotes convener

- ^{*}**Weldeghebriel, M. F.**, Lowenstein, T. K., Murphy, J. G., Jurikova, H., Rae, J. W. B., Niespolo, E. M., Higgins, J. A. A 550-million-year record of the concentration and isotopic composition of lithium in seawater. Goldschmidt 2026, Montreal, Canada. <https://conf.goldschmidt.info/goldschmidt/2026/meetingapp.cgi/Paper/35051>
- Weldeghebriel, M. F.**, Fairuz, I., Higgins, J. A., Niespolo, E. M. High-resolution 2D impurity mapping of million-year-old Antarctic ice by cryo-LA-ICP-MS. Goldschmidt 2026, Montreal, Canada. <https://conf.goldschmidt.info/goldschmidt/2026/meetingapp.cgi/Paper/36231>
- [#]**Weldeghebriel, M. F.**, Lowenstein, T. K., Murphy, J. G., Jurikova, H., Rae, J. W.B., Niespolo, E. M., Higgins, J. A. Secular variations in Phanerozoic seawater lithium and links to Earth's climate states and the carbon–silicate cycle. Gordon Research Conference, 2025, Maine, USA. <https://www.grc.org/geochronology-grs-conference/2025/>
- ^{*}**Weldeghebriel, M. F.**, Xia, Z., Li, W., Lonsdale, M., Smith, E. F., Blättler, C., Nadeau, M., Murphy, J. G., Niespolo, E. M., Higgins, J. A. A 3-billion-year lithium isotope record from marine dolostones. Goldschmidt 2025, Prague, Czech Republic. <https://goldschmidtabstracts.info/abstracts/abstractView?doi=10.7185/gold2025.27539>
- [#]**Weldeghebriel, M. F.**, Lowenstein, T. K., Murphy, J. G., Jurikova, H., Rae, J. W.B., Niespolo, E. M., Higgins, J. A. Secular variation of lithium concentration and isotopic composition of Phanerozoic and Neoproterozoic seawater. Goldschmidt 2024, Chicago, USA. <https://conf.goldschmidt.info/goldschmidt/2024/meetingapp.cgi/Paper/23022>

- *Weldeghebriel, M. F.,** Lowenstein, T. K., Higgins, J. A. Variability in strontium and lithium composition of ancient seawater from fluid inclusions in halite. EGU General Assembly 2024, Vienna, Austria.
<https://meetingorganizer.copernicus.org/EGU24/EGU24-13908.html>
- Weldeghebriel, M. F.,** et al. Reconstructing the secular evolution of the lithium isotope composition of Phanerozoic and Neoproterozoic seawater from marine halite. GSA Annual Meeting 2023, Pittsburgh, USA.
<https://gsa.confex.com/gsa/2023AM/meetingapp.cgi/Paper/395704>
- #Weldeghebriel, M. F.,** Lowenstein, T. K., Higgins, J. A. Variability in strontium and lithium concentration of ancient seawater from fluid inclusions in halite. Chapman Conference on Hydrothermal Systems, 2023, Agros, Cyprus. <https://www.agu.org/Chapman-Hydrothermal-Circulation-Seawater-Chemistry>
- Weldeghebriel, M. F.,** Lowenstein, T. K., Sirota, I., Eyal, H., Mor, Z., Lensky, N. Coupling halite deposition in the Dead Sea to environmental conditions. AGU Fall Meeting 2021, New Orleans, USA. <https://studio.m-anage.com/agu/fm21/meetingapp.cgi/Paper/879177>
- Weldeghebriel, M. F.,** Lowenstein, T. K. Secular variability of Phanerozoic seawater strontium: evidence from fluid inclusions in marine halite. AGU Fall Meeting 2019, San Francisco, USA.
<https://ui.adsabs.harvard.edu/abs/2019AGUFMPP51E1411W/abstract>
- Weldeghebriel, M. F.,** Lowenstein, T. K., García-Veigas, J., Cendon, D. I. Ca-SO₄ crossover points in Phanerozoic seawater: Evidence from fluid inclusions in halite. AGU Fall Meeting 2019, San Francisco, USA.
<https://studio.m-anage.com/agu/fm21/meetingapp.cgi/Paper/879935>
- Weldeghebriel, M. F.,** Lowenstein, T. K., Demicco, R. V., Graney, J. R., Collins, D., Veigas, J. G., ... & Sendula, E., 2019. Seafloor hydrothermal systems control seawater chemistry: evidence from fluid inclusions in halite. GSA Annual Meeting 2019, Phoenix, USA. <https://gsa.confex.com/gsa/2019AM/meetingapp.cgi/Paper/335157>
- Weldeghebriel, M. F.,** Lowenstein, T. K. Major, minor and trace element evolution of Phanerozoic seawater. GSA Northeastern Section 53rd Annual Meeting, 2018, Burlington, Vermont, USA.
<https://gsa.confex.com/gsa/2018NE/meetingapp.cgi/Paper/311328>
- Weldeghebriel, M. F.** Sedimentation and lithostratigraphic succession of evaporites in the Danakil Depression, Colluli, Eritrea. 3rd World Young Earth Scientists Congress, 2014, Dar es Salaam, Tanzania. [T2-18]
https://www.academia.edu/11285678/Abstract_volume_of_YES_congress_2014_at_Dar_es_Salaam_Tanzania
- Weldeghebriel, M. F.,** Kiflai, M. E., 2014. A proposed-detailed investigation on petrography, geochemistry, and geochronology of Neoproterozoic granitoid rocks in Dekemhare Area, Eritrea. 3rd World Young Earth Scientists Congress, 2014, Dar es Salaam, Tanzania. [T2-17]
https://www.academia.edu/11285678/Abstract_volume_of_YES_congress_2014_at_Dar_es_Salaam_Tanzania

TEACHING

- Binghamton University, Graduate Teaching Assistant (2016–2021): laboratory sections of 15–30 students each semester for five years in Geochemistry (GEOL 470/570), Planet Earth (GEOL 111), Earth's Interior (GEOL 214), Climate and Paleoclimate (GEOL 414/514), and Geology of the Solar System (GEOL 102).
- Eritrea Institute of Technology, Lecturer and Teaching Assistant (2011–2015): General Geology, Geological Map Interpretation, Rock-Forming Minerals, Introduction to Geophysics, and Field Mapping; led eight undergraduate field camps for more than 100 students.
- Mentoring: trained an undergraduate intern, four graduate students, a postdoctoral researcher, and a laboratory technician at Princeton and Binghamton in LA-ICP-MS and lithium-isotope analytical methods; supervised more than 40 undergraduate field projects and theses at the Eritrea Institute of Technology.

TALKS

- Secular variability in major, trace element, and isotopic composition of Phanerozoic and Neoproterozoic seawater: Implications for reconstructing drivers of seawater secular variability and climate ([Environmental Geology and Geochemistry Seminar](#), Princeton University, Nov. 9, 2023).
- Secular variability in major, trace element, and isotopic composition of Phanerozoic and Neoproterozoic seawater: Implications for reconstructing drivers of seawater secular variability and climate ([AOES/GEOL Seminar](#), George Mason University, Oct. 26, 2023, virtual).

ACADEMIC AND PROFESSIONAL SERVICE AND OUTREACH

- Coordinator, Environmental Geology and Geochemistry Seminar (EGGS) Lecture Series, Department of Geosciences, Princeton University (2025–present).
- **Session Convener**, “Seawater Chemistry Through Time: Reconstructions, Mechanisms, and Feedbacks,” Goldschmidt 2026.
- **Session Convener**, “The Evolution of Seawater Chemistry Through Time,” Goldschmidt 2025.
- **Session Convener**, “The Carbon Cycle as Reflected in Ocean Chemistry and Marine Records,” EGU General Assembly 2024.
- **Peer reviewer** for *Nature Communications*, *Geology*, *Earth-Science Reviews*, *Journal of Geochemical Exploration*, and *Geological Quarterly*.
- **Community service**, Mathematics and Life Skills Teacher, Shambuko Junior and Secondary School, 2008–2009.

FIELD AND INDUSTRY EXPERIENCE

- Lithium Americas Corp., Nevada, USA, Exploration Intern (Jun–Sep 2021): field exploration of lithium-bearing sedimentary basins at 25+ sites across six U.S. states; 500+ samples collected.
- South Boulder Mines Ltd., Eritrea, Exploration Intern (Jun 2010–Mar 2011): evaporite mapping and potash drill-core logging at the Colluli Potash Project, Danakil Depression.
- Dead Sea, Israel: coring and sedimentology of modern halite as an analog to deep hypersaline basins (Dec–Jan 2019/20).
- Afar Rift and Danakil Block, Eritrea: GPS geodetic surveys and broadband seismometer deployments, including monitoring of the 2011 Nabro eruption (2012–2014).